

Hanjia Lyu

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Research Areas

Computational Social Science, Multimodal Learning, Graph Machine Learning

Education

Ph.D. in Computer Science, University of Rochester 2021–2026

Advisor: Jiebo Luo

M.S. in Data Science, University of Rochester 2019–2020

B.A. in Economics, Fudan University 2013–2017

Professional Experience

Singapore Management University *Singapore*
Assistant Professor, Department of Computing and Information Systems July 2026–

Snap *Remote*
Research Scientist Intern May 2025–Apr 2026

Meta *Menlo Park, CA*
Research Scientist Intern Jun 2024–Jan 2025
Apr–Aug 2023
May–Sep 2022

Wyze Labs *Remote*
Research Scientist Intern Jan–May 2022

University of Rochester Medical Center *Rochester, NY*
Research Assistant Jan–May 2021

Recognition

◇ Google PhD Fellowship 2024

◇ Donald M. and Janet C. Barnard Fellowship 2023

Conference/Journal Organizing Experience

Area Chair

- ◇ Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL) 2025
- ◇ Conference on Empirical Methods in Natural Language Processing (EMNLP) 2024

Senior Program Committee

- ◇ International AAAI Conference on Web and Social Media (ICWSM) 2026–2027

Guest Editor for Journals

- ◇ Frontiers in Public Health
- ◇ ACM Transactions on Multimedia Computing, Communications, and Applications
 - Special Issue on Multimodal Video Understanding and Analysis with Foundation Models

Organizing Committee

- ◇ 1st Multimodal Learning for Social Good (MML4SG) workshop at the IEEE International Conference on Multimedia and Expo (ICME) 2024
- ◇ 1st Visual-Language Alignment in Text-Guided Multi-Modal Generation (TMMG) workshop at the IEEE International Conference on Multimedia and Expo (ICME) 2024

Session Chair

- ◇ International AAAI Conference on Web and Social Media (ICWSM) 2026
- ◇ ACM International Conference on Information and Knowledge Management (CIKM) 2024

Publications

Refereed Journal Papers

1. **Hanjia Lyu***, Jinfa Huang*, Daoan Zhang*, Yongsheng Yu*, Xinyi Mou*, Jinsheng Pan, Zhengyuan Yang, Zhongyu Wei, and Jiebo Luo. GPT-4V(ision) as a social media

analysis engine. *ACM Trans. Intell. Syst. Technol.*, 16(3), April 2025. ISSN 2157-6904. doi: 10.1145/3709005. URL <https://doi.org/10.1145/3709005>

→ Presented at: *Synchronous Reading Club*

2. Weihong Qi, Jinsheng Pan, **Hanjia Lyu**, and Jiebo Luo. Excitements and concerns in the post-ChatGPT era: Deciphering public perception of AI through social media analysis. *Telematics and Informatics*, 2024. URL <https://doi.org/10.1016/j.tele.2024.102158>

→ Media coverage: *UofR News Center, Futurity*

3. Junyu Chen, Jie An, **Hanjia Lyu**, Christopher Kanan, and Jiebo Luo. Learning to evaluate the artness of AI-generated images. *IEEE Transactions on Multimedia*, 2024. URL <https://doi.org/10.1109/TMM.2024.3410672>
4. **Hanjia Lyu**, Aarsal Imtiaz, Yufei Zhao, and Jiebo Luo. Human behavior in the time of COVID-19: Learning from big data. *Frontiers in Big Data*, 6:1099182, 2023. URL <https://doi.org/10.3389/fdata.2023.1099182>
5. Yangxin Fan, **Hanjia Lyu**, Jin Xiao, and Jiebo Luo. American Twitter users revealed social determinant-related oral health disparities amid the COVID-19 pandemic. *Quintessence International*, 54(1), 2023. URL <https://doi.org/10.3290/j.qi.b3479979>
6. **Hanjia Lyu**, Zihe Zheng, and Jiebo Luo. Misinformation versus facts: Understanding the influence of news regarding COVID-19 vaccines on vaccine uptake. *Health Data Science*, 2022. URL <https://doi.org/10.34133/2022/9858292>

→ Media coverage: *Medical Economics, News Medical, News Wise, Medical Xpress*

→ Presented at: *Great Lakes Science Boot Camp for Librarians, CSUG Research Panel*

7. **Hanjia Lyu**, Yangxin Fan, Ziyu Xiong, Mayya Komisarchik, and Jiebo Luo. Understanding public opinion toward the #StopAsianHate movement and the relation with racially motivated hate crimes in the US. *IEEE Transactions on Computational Social Systems*, 2021. URL <https://doi.org/10.1109/TCSS.2021.3136858>

→ Media coverage: *UofR News Center, Rochester Beacon, Futurity*

8. **Hanjia Lyu**, Junda Wang, Wei Wu, Viet Duong, Xiyang Zhang, Timothy D. Dye, and Jiebo Luo. Social media study of public opinions on potential COVID-19 vaccines: informing dissent, disparities, and dissemination. *Intelligent Medicine*, 2(1):1–12, 2022. ISSN 2667-1026. doi: <https://doi.org/10.1016/j.imed.2021.08.001>. URL <https://www.sciencedirect.com/science/article/pii/S266710262100036X>

→ Media coverage: *News Medical*

→ Presented at: *Great Lakes Science Boot Camp for Librarians, CSUG Research Panel*

9. Ziyu Xiong, Pin Li, **Hanjia Lyu**, and Jiebo Luo. Social media opinions on working from home in the United States during the COVID-19 pandemic: Observational study. *JMIR medical informatics*, 9(7):e29195, 2021. URL <https://doi.org/10.2196/29195>
10. Wei Wu, **Hanjia Lyu**, and Jiebo Luo. Characterizing discourse about COVID-19 vaccines: A Reddit version of the pandemic story. *Health Data Science*, 2021. URL <https://doi.org/10.34133%2F2021%2F9837856>
11. Xupin Zhang, **Hanjia Lyu**, and Jiebo Luo. What contributes to a crowdfunding campaign’s success? evidence and analyses from GoFundMe data. *Journal of Social Computing*, 2(2):183–192, 2021. URL <https://doi.org/10.23919/JSC.2021.0010>

→ Media coverage: *The New York Times, The Wall Street Journal*

12. Xiyang Zhang, Yu Wang, **Hanjia Lyu**, Yipeng Zhang, Yubao Liu, and Jiebo Luo. The influence of COVID-19 on the well-being of people: Big data methods for capturing the well-being of working adults and protective factors nationwide. *Frontiers in Psychology*, 12:681091, 2021. URL <https://doi.org/10.3389%2Ffpsyg.2021.681091>
13. Yipeng Zhang, **Hanjia Lyu**, Yubao Liu, Xiyang Zhang, Yu Wang, and Jiebo Luo. Monitoring depression trends on Twitter during the COVID-19 pandemic: Observational study. *JMIR infodemiology*, 1(1):e26769, 2021. URL <https://doi.org/10.2196/26769>

→ Media coverage: *Tech Xplore, UofR News Center*

14. Siqing Cao, **Hanjia Lyu**, and Xian Xu. Insurtech development: Evidence from Chinese media reports. *Technological Forecasting and Social Change*, 161:120277, 2020. URL <https://doi.org/10.1016/j.techfore.2020.120277>
15. **Hanjia Lyu**, Long Chen, Yu Wang, and Jiebo Luo. Sense and sensibility: Characterizing social media users regarding the use of controversial terms for COVID-19. *IEEE Transactions on Big Data*, 7(6):952–960, 2020. URL <https://doi.org/10.1109%2FTBDATA.2020.2996401>

→ Media coverage: *IEEE Spectrum, Tech Xplore, UofR News Center*

→ Presented at: *Experience Rochester, CSUG Research Panel*

Refereed Conference Papers

1. **Hanjia Lyu**. Building human-aligned AI through representation augmentation and structuring. In *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining*, WSDM ’26, New York, NY, USA, 2026. Association for Computing Machinery. ISBN 9798400722929. doi: 10.1145/3773966.3784981. URL <https://doi.org/10.1145/3773966.3784981>

→ Will be presented at: WSDM 2026

2. **Hanjia Lyu**, Jiebo Luo, Jian Kang, and Allison Koenecke. Characterizing bias: Benchmarking large language models in simplified versus traditional Chinese. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*, FAccT '25, pp. 2815–2846, New York, NY, USA, 2025. Association for Computing Machinery. ISBN 9798400714825. doi: 10.1145/3715275.3732182. URL <https://doi.org/10.1145/3715275.3732182>

→ Presented at: FAccT 2025, IC2S2 2025 (Parallel Talk)

3. Chenwei Lin, **Hanjia Lyu**, Xian Xu, and Jiebo Luo. INS-MMBench: A comprehensive benchmark for evaluating lvlms’ performance in insurance. In *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*, Honolulu, HI, USA, October 2025. IEEE Computer Society. URL <https://arxiv.org/pdf/2406.09105>
4. Sreejato Chatterjee, Linh Tran, Quoc Duy Nguyen, Roni Kirson, Drue Hamlin, Harvest Aquino, **Hanjia Lyu**, Jiebo Luo, and Timothy Dye. Assessing Historical Structural Oppression Worldwide via Rule-Guided Prompting of Large Language Models. In *2025 IEEE International Conference on Big Data (BigData)*, Macau, China, December 2025. IEEE Computer Society. URL <https://arxiv.org/pdf/2509.15216>
5. Yawen Zheng, **Hanjia Lyu**, and Jiebo Luo. Irony in Emojis: A Comparative Study of Human and LLM Interpretation. In *2025 IEEE International Conference on Big Data (BigData)*, Macau, China, December 2025. IEEE Computer Society. URL <https://arxiv.org/pdf/2501.11241>
6. Yu Xia, Subhojyoti Mukherjee, Zhouhang Xie, Junda Wu, Xintong Li, Ryan Aponte, **Hanjia Lyu**, Joe Barrow, Hongjie Chen, Franck Dernoncourt, et al. From selection to generation: A survey of llm-based active learning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Vienna, Austria, 2025. Association for Computational Linguistics. URL <https://arxiv.org/pdf/2502.11767>
7. Dang Nguyen, Jian Chen, Yu Wang, Gang Wu, Namyong Park, Zhengmian Hu, **Hanjia Lyu**, Junda Wu, Ryan Aponte, Yu Xia, et al. Gui agents: A survey. In *Findings of the Association for Computational Linguistics: ACL 2025*, Vienna, Austria, 2025. Association for Computational Linguistics. URL <https://arxiv.org/pdf/2412.13501>
8. Xiaofei Zhou, **Hanjia Lyu**, Yuxin Sa, Mengfan Li, Advait Sarkar, Jiebo Luo, Michael Daley, and Zhen Bai. Empower K-12 teachers to create ml-supported scientific inquiries. In *International Conference on Artificial Intelligence in Education, AIED*, Palermo, Italy, July, 2025

→ Presented at: AIED 2025 (**Best Paper Nominee**)

9. Weihong Qi, **Hanjia Lyu**, and Jiebo Luo. Representation bias in political sample simulations with large language models. In *Companion Proceedings of the ACM on Web*

Conference 2025, WWW '25, pp. 1264–1267, New York, NY, USA, 2025. Association for Computing Machinery. ISBN 9798400713316. doi: 10.1145/3701716.3715591. URL <https://doi.org/10.1145/3701716.3715591>

10. Jinfa Huang*, Jinsheng Pan*, Zhongwei Wan, **Hanjia Lyu**, and Jiebo Luo. Evolver: Chain-of-evolution prompting to boost large multimodal models for hateful meme detection. In *The 31st International Conference on Computational Linguistics*, Abu Dhabi, UAE, 2025. URL <https://arxiv.org/pdf/2407.21004>
11. **Hanjia Lyu**, Song Jiang, Hanqing Zeng, Yinglong Xia, Qifan Wang, Si Zhang, Ren Chen, Chris Leung, Jiajie Tang, and Jiebo Luo. LLM-Rec: Personalized recommendation via prompting large language models. In Kevin Duh, Helena Gomez, and Steven Bethard (eds.), *Findings of the Association for Computational Linguistics: NAACL 2024*, pp. 583–612, Mexico City, Mexico, June 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-naacl.39. URL <https://aclanthology.org/2024.findings-naacl.39/>
 → *Presented at: NAACL 2024, DEFIRST Reading Group at Mila x Vector Institute*
12. Hanqing Zeng*, **Hanjia Lyu***, Diyi Hu, Yinglong Xia, and Jiebo Luo. Mixture of weak & strong experts on graphs. In *The Twelfth International Conference on Learning Representations, ICLR*, Vienna, Austria, May, 2024. URL <https://openreview.net/pdf?id=wYvuY60SdD>
13. **Hanjia Lyu***, Jinsheng Pan*, Zichen Wang*, and Jiebo Luo. Computational assessment of hyperpartisanship in news titles. *Proceedings of the International AAAI Conference on Web and Social Media*, 18(1):999–1012, May 2024. doi: 10.1609/icwsm.v18i1.31368. URL <https://ojs.aaai.org/index.php/ICWSM/article/view/31368>
 → *Media coverage: UofR News Center, Study Finds, ScienceBlog, EurekAlert!*
14. **Hanjia Lyu***, Weihong Qi*, Zhongyu Wei, and Jiebo Luo. Human vs. LMMs: Exploring the discrepancy in emoji interpretation and usage in digital communication. In *AAAI International Conference on Web and Social Media, ICWSM*, Buffalo, NY, USA, June, 2024. URL <https://arxiv.org/pdf/2401.08212.pdf>
15. **Hanjia Lyu**. Towards advancing text-based user and item representation in personalized recommendation. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management, CIKM '24*, New York, NY, USA, 2024. URL <https://doi.org/10.1145/3627673.3680270>
16. Zhongyi Qiu, Kangyi Qiu, **Hanjia Lyu**, Wei Xiong, and Jiebo Luo. Semantics Preserving Emoji Recommendation with Large Language Models. In *2024 IEEE International Conference on Big Data (BigData)*, pp. 7131–7140, Los Alamitos, CA, USA, December 2024. IEEE Computer Society. doi: 10.1109/BigData62323.2024.10825696. URL <https://doi.ieeecomputersociety.org/10.1109/BigData62323.2024.10825696>

17. Yiyao Tao*, Hengyu Zhang*, Bali Dey, Selenge Tulga, **Hanjia Lyu**, and Jiebo Luo. In the eyes of the bystander: Are the stances on different conflicts correlated? In *Special Session on Intelligent Data Mining, IEEE Big Data Conference*, Washington DC, USA, December, 2024. URL <https://arxiv.org/pdf/2401.08733>
18. Junda Wang, Weijian Li, Han Wang, **Hanjia Lyu**, Caroline Thirukumaran, Addisu Mesfin, Hong Yu, and Jiebo Luo. Crtre: Causal rule generation with target trial emulation framework. In *IEEE Big Data Conference*, Washington DC, USA, December, 2024. URL <https://arxiv.org/pdf/2209.14975.pdf>
19. Daoan Zhang, Junming Yang, **Hanjia Lyu**, Zijian Jin, Yuan Yao, Mingkai Chen, and Jiebo Luo. A benchmark and chain-of-thought prompting strategy for large multi-modal models with multiple image inputs. In *International Conference on Pattern Recognition, ICPR*, Kolkata, India, December, 2024. URL <https://arxiv.org/pdf/2401.02582.pdf>
20. Xinnong Zhang, Haoyu Kuang, Xinyi Mou, **Hanjia Lyu**, Kun Wu, Siming Chen, Jiebo Luo, Xuanjing Huang, and Zhongyu Wei. SoMeLVLM: A large vision language model for social media processing. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar (eds.), *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 2366–2389, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.140. URL <https://aclanthology.org/2024.findings-acl.140/>
21. Junyu Chen, Jie An, **Hanjia Lyu**, Christopher Kanan, and Jiebo Luo. Holistic visual-textual sentiment analysis with prior models. In *2024 IEEE 7th International Conference on Multimedia Information Processing and Retrieval (MIPR)*, San Jose, CA, USA, August, 2024. URL <https://doi.ieeecomputersociety.org/10.1109/MIPR62202.2024.00037>
22. Xiaofei Zhou, Jingwan Tang, **Hanjia Lyu**, Xinyi Liu, Zhenhao Zhang, Lichen Qin, Fiona Au, Advait Sarkar, and Zhen Bai. Creating an authoring tool for K-12 teachers to design ML-supported scientific inquiry learning. In *Late Breaking Papers: ACM Conference on Human Factors in Computing Systems, CHI*, Honolulu, HI, USA, May, 2024. URL <https://doi.org/10.1145/3613905.3650762>
23. Xinyi Mou, Zejun Li, **Hanjia Lyu**, Jiebo Luo, and Zhongyu Wei. Unifying local and global knowledge: Empowering large language models as political experts with knowledge graphs. In *Proceedings of the ACM Web Conference 2024, WWW '24*, pp. 2603–2614, New York, NY, USA, 2024. Association for Computing Machinery. ISBN 9798400701719. doi: 10.1145/3589334.3645616. URL <https://doi.org/10.1145/3589334.3645616>
24. **Hanjia Lyu***, Kevin C. Kapcio*, Kyle C. Purrmann, Christian G. Peyre, Carolyn E. Jones, Michal J. Lada, and Jiebo Luo. Applying machine learning to predict esophageal cancer recurrence after esophagectomy. In *IEEE International Conference on Digital Health, ICDH*, Chicago, IL, USA, July, 2023. URL <https://doi.org/10.1109/ICDH60066.2023.00044>

25. Jinsheng Pan, Zichen Wang, Weihong Qi, **Hanjia Lyu**, and Jiebo Luo. Understanding divergent framing of the supreme court controversies: Social media vs. news outlets. In *Special Session on Intelligent Data Mining, IEEE Big Data Conference*, Sorrento, Italy, December, 2023. URL <https://doi.ieeecomputersociety.org/10.1109/BigData59044.2023.10386532>
26. Enting Zhou*, Yurong Liu*, **Hanjia Lyu**, and Jiebo Luo. A fine-grained analysis of public opinion toward Chinese technology companies on Reddit. In *Special Session on Intelligent Data Mining, IEEE Big Data Conference*, Sorrento, Italy, December, 2023. URL <https://doi.ieeecomputersociety.org/10.1109/BigData59044.2023.10386804>
27. Mohammad Mahdi Kamani, Yuhang Yao, **Hanjia Lyu**, Zhongwei Cheng, Lin Chen, Liangju Li, Carlee Joe-Wong, and Jiebo Luo. Wyze rule: Federated rule dataset for rule recommendation benchmarking. In *Thirty-seventh Conference on Neural Information Processing Systems, NeurIPS*, New Orleans, LA, USA, December, 2023. URL <https://openreview.net/pdf?id=qynH28Y4xE>
28. Edinam Kofi Klutse, Samuel Nuamah-Amoabeng, **Hanjia Lyu**, and Jiebo Luo. Dismantling hate: Understanding hate speech trends against nba athletes. In *International Conference on Social Computing, Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation, SBP-BRiMS*, Pittsburgh, PA, USA, September, 2023. URL https://doi.org/10.1007/978-3-031-43129-6_8
29. Jingyang Lin, Junyu Chen, **Hanjia Lyu**, Igor Khodak, Divya Chhabra, Colby L Day Richardson, Irina Prelipcean, Andrew M Dylag, and Jiebo Luo. Predicting adverse neonatal outcomes for preterm neonates with multi-task learning. In *IEEE International Conference on Digital Health, ICDH*, Chicago, IL, USA, July, 2023. URL <https://doi.org/10.1109/ICDH60066.2023.00045>
30. Junyu Chen, Jie An, **Hanjia Lyu**, and Jiebo Luo. How art-like are AI-generated images? an exploratory study. In *Proceedings of the 1st International Workshop on Multimedia Content Generation and Evaluation: New Methods and Practice*, McGE '23, pp. 19–22, New York, NY, USA, 2023. Association for Computing Machinery. ISBN 9798400702785. doi: 10.1145/3607541.3616825. URL <https://doi.org/10.1145/3607541.3616825>
31. Jinsheng Pan*, Weihong Qi*, Zichen Wang, **Hanjia Lyu**, and Jiebo Luo. Bias or diversity? Unraveling fine-grained thematic discrepancy in U.S. news headlines. In *Workshop on News Media and Computational Journalism, MEDIATE, AAAI International Conference on Web and Social Media, ICWSM*, Limassol, Cyprus, June, 2023. URL https://workshop-proceedings.icwsm.org/pdf/2023_25.pdf
 → Media coverage: *UofR News Center, Study Finds, ScienceBlog, EurekAlert!*
32. Xiaofei Zhou, **Hanjia Lyu**, Jiebo Luo, and Zhen Bai. ML-SD modeling: How machine learning can support scientific discovery learning for K-12 stem education. In *Artificial*

Intelligence for Education Workshop AI4Edu, AAAI Conference on Artificial Intelligence, AAAI, Washington DC, USA, February, 2023. URL <https://par.nsf.gov/servlets/purl/10427044>

33. Sajid Hussain Rafi Ahamed, Shahid Shakil, **Hanjia Lyu**, Xinpeng Zhang, and Jiebo Luo. Doctors vs. nurses: Understanding the great divide in vaccine hesitancy among healthcare workers. In *Special Session on Intelligent Data Mining, IEEE Big Data Conference*, Osaka, Japan, December, 2022. URL <https://doi.ieeecomputersociety.org/10.1109/BigData55660.2022.10020853>
34. **Hanjia Lyu** and Jiebo Luo. Understanding political polarization via jointly modeling users, connections and multimodal contents on heterogeneous graphs. In *Proceedings of the 30th ACM International Conference on Multimedia*, MM '22, pp. 4072–4082, New York, NY, USA, 2022. Association for Computing Machinery. ISBN 9781450392037. doi: 10.1145/3503161.3547898. URL <https://doi.org/10.1145/3503161.3547898>
 → *Presented at: MM 2022, Meta, Western New York Image and Signal Processing Workshop*
35. Wei Zhu, Ziheng Zheng, Haitian Zheng, **Hanjia Lyu**, and Jiebo Luo. Learning to aggregate and refine noisy labels for visual sentiment analysis. In *International Conference on Pattern Recognition, ICPR*, Montréal, Canada, August, 2022. URL <https://doi.ieeecomputersociety.org/10.1109/ICPR56361.2022.9956328>
36. Xiaofei Zhou, Jingwan Tang, Beilei Guo, **Hanjia Lyu**, and Jiebo Luo. Challenges and design opportunities in data analysis for ml-empowered scientific inquiry – insights from a teacher professional development study. In *International Conference of the Learning Sciences, ICLS*, Virtual, June, 2022. URL <https://repository.isls.org/handle/1/9077>
37. Aarsal Imtiaz, Danish Khan, **Hanjia Lyu**, and Jiebo Luo. Taking sides: Public opinion over the Israel-Palestine conflict in 2021. In *International Workshop on Social Sensing, SocialSens: Special Edition on Belief Dynamics, AAAI International Conference on Web and Social Media, ICWSM*, Atlanta, GA, USA and online, June, 2022. URL https://workshop-proceedings.icwsml.org/pdf/2022_30.pdf
38. Yichi Qian, Qiyi Shan, **Hanjia Lyu**, and Jiebo Luo. Look behind the censorship: Reposting-user characterization and muted-topic restoration. In *International Workshop on Social Sensing, SocialSens: Special Edition on Belief Dynamics, AAAI International Conference on Web and Social Media, ICWSM*, Atlanta, GA, USA and online, June, 2022. URL https://workshop-proceedings.icwsml.org/pdf/2022_29.pdf
39. Tanqiu Jiang, Sidhant Bendre, **Hanjia Lyu**, and Jiebo Luo. From static to dynamic prediction: Wildfire risk assessment based on multiple environmental factors. In *Special Session on Intelligent Data Mining, IEEE Big Data Conference*, Virtual, December, 2021. URL <https://doi.org/10.1109/BigData52589.2021.9672044>

40. Xupin Zhang, **Hanjia Lyu**, and Jiebo Luo. Understanding the hoarding behaviors during the COVID-19 pandemic using large scale social media data. In *Special Session on Intelligent Data Mining, IEEE Big Data Conference*, Virtual, December, 2021. URL <https://doi.ieeecomputersociety.org/10.1109/BigData52589.2021.9671533>

→ Media coverage: *Tech Xplore*, *UofR News Center*
41. Long Chen, **Hanjia Lyu**, Tongyu Yang, Yu Wang, and Jiebo Luo. Fine-grained analysis of the use of neutral and controversial terms for COVID-19 on social media. In *International Conference on Social Computing, Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation, SBP-BRiMS*, Virtual, July, 2021. URL https://doi.org/10.1007/978-3-030-80387-2_6

→ Media coverage: **IEEE Spectrum**, *Tech Xplore*, *UofR News Center*
42. Karan Vombatkere, **Hanjia Lyu**, and Jiebo Luo. How political is the spread of COVID-19 in the United States? An analysis using transportation and weather data. In *International Conference on Social Computing, Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation, SBP-BRiMS*, Virtual, July, 2021. URL https://doi.org/10.1007/978-3-030-80387-2_2

Abstracts & Conference Presentations

1. Shazia Siddiqi, Tiffany Panko, Wyatt Hall, Lisette Alcantara, **Hanjia Lyu**, Jiebo Luo, Eva Pressman, and Timothy Dye. Multi-gendered, global analysis of reproductive intention among deaf, hard of hearing, and hearing populations. In *American Public Health Association (APHA) Annual Meeting and Expo*, Washington, D.C., 2025. URL <https://apha.confex.com/apha/2025/meetingapp.cgi/Paper/576909>. Conference abstract
2. Timothy Dye, Brooke Levandowski, Monica Barbosu, Lisette Alcantara, Esteban Avendaño, Jiebo Luo, **Hanjia Lyu**, Saloni Sharma, Nazifa Shinwarie, Shazia Siddiqi, Ivelisse Rivera, and Eva Pressman. Potential covid-19 impacts of lived experience on ecosystems of parenting families worldwide: Results of a multilingual global survey from 168 countries. In *American Public Health Association (APHA) Annual Meeting and Expo*, Washington, D.C., 2025. URL <https://apha.confex.com/apha/2025/meetingapp.cgi/Paper/578291>. Conference abstract
3. Timothe Dye, Lisette Alcántara, **Hanjia Lyu**, Shazia Siddiqi, Saloni Sharma, Eva Pressman, and Jiebo Luo. Oppression, COVID vaccination, and vaccine sentiments in a global sample. *Annals of Epidemiology*, 75:82, 2022. URL <https://www.sciencedirect.com/science/article/am/pii/S1047279722001983>. Conference abstract

Teaching Experience

Guest Lecturer

University of Rochester CSC 511: Large Language Models	Nov 2024
University of Rochester CSC 440: Data Mining	Sep 2023

Teaching Assistant

University of Rochester CSC 480: Computer Models & Limitations	Spring 2023
CSC 486: Computational Complexity	Spring 2023
CSC 440: Data Mining	Fall 2022
CSC 446: Machine Learning	Spring 2022
CSC 440: Data Mining	Fall 2021
CSC 440: Data Mining	Fall 2020

Talks & Publicity

◇ Presentation at the ACM International Conference on Web Search and Data Mining (WSDM)	Feb 2026
◇ Presentation at the ACM Conference on Fairness, Accountability, and Transparency (FAccT)	Jun 2025
◇ Invited talk at Meta	Jun 2025
◇ Presentation at the ACM International Conference on Information and Knowledge Management (CIKM)	Oct 2024
◇ Presentation at the IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)	Aug 2024
◇ Presentation at the International AAAI Conference on Web and Social Media (ICWSM)	Jun 2024
◇ Presentation at the Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)	Jun 2024
◇ Presentation at the International Conference on Learning Representations (ICLR)	May 2024
◇ Invited talk at Data Science Snack Series: Internship Panel, University of Rochester	Feb 2024
◇ Invited talk at Synchronous Reading Club	Jan 2024

◇ Invited talk at Fudan University	Nov 2023
◇ Invited talk at DEFIRST Reading Group at Mila x Vector Institute	Oct 2023
◇ Presentation at the IEEE International Conference on Digital Health (ICDH)	Jul 2023
◇ Presentation at the AS&E Graduate Research Symposium, University of Rochester	Mar 2023
◇ Invited talk at the Western New York Image and Signal Processing Workshop	Nov 2022
◇ Invited talk at Meta	Aug 2022
◇ Invited talk at the Great Lakes Science Boot Camp for Librarians	Jun 2022
◇ Invited talk at Fudan-UC Center on Contemporary China, UCSD	May 2022
◇ Invited talk at InsurTech Innovation: UK & China as Hotspots of the Global World	Apr 2022
◇ Invited talk at CSUG Research Panel, University of Rochester	Oct 2021
◇ Presentation at the International Conference on Social Computing, Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation (SBP-BRiMS)	Jul 2021
◇ Invited talk at Experience Rochester, University of Rochester	Sep 2020